Name: Jeffrey Submitted 2/23

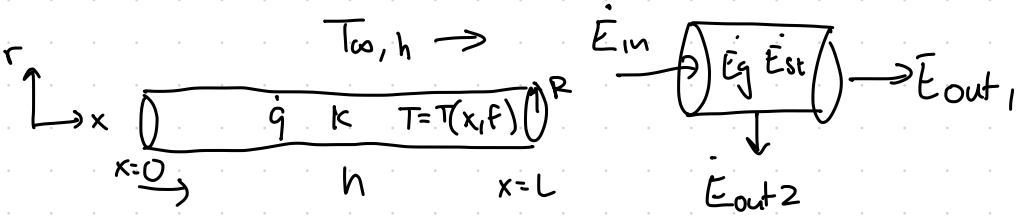
PS#3 Due 2/24

Acknowledgement: None: Course notes

Given R, $\dot{q}$, T_∞ ambient, h T is func $T(x,t)$ k ρ C_p

Find a) Energy balance equations b) Diff eq c) BCs & ICs

Schematic:



Assumptions: No radiation, k ρ C_p constant, R small 1D,

Analysis: Balance energy transfer

a) $\dot{E}_{in} + \dot{E}_g - \dot{E}_{out_1} - \dot{E}_{out2} = \dot{E}_{st}$

$\dot{E}_{in_1} = q_x = -kA_c\frac{dT}{dx}$ $\quad q_{x+dx} = q_x + \frac{dq_x}{dx}dx = \dot{E}_{out_2}$ $\quad \dot{E}_{st} = \rho C_p A\frac{\partial T}{\partial t}$

$q_{conv} = hA_s(T-T_\infty) = \dot{E}_{out_1}$ $\quad \dot{E}_g = \dot{q}A\,dx$

$$A\dot{q}\,dx + k\pi R^2\frac{dT}{dx} - q_x - \frac{dq_x}{dx}dx - 2\pi Rh\,dx(T-T_\infty) = \dot{E}_{st}$$

b) $A\dot{q}\,dx - 2\pi Rh\,dx(T-T_\infty) - \frac{\partial q_x}{\partial x}dx = \dot{E}_{st}$

$$\dot{q}(\pi R^2)\,dx - 2\pi Rh(T-T_\infty)\,dx - \frac{\partial}{\partial x}\left(+kA\frac{\partial T}{\partial x}\right)dx = \rho C_p(A\,dx)\frac{\partial T}{\partial t} = \rho C_p\pi R^2\,dx\frac{\partial T}{\partial t}$$

$$\dot{q}(\pi R^2) - 2\pi Rh(T-T_\infty) - \frac{\partial}{\partial x}\left(k\pi R^2\frac{\partial T}{\partial x}\right) = \rho C_p\pi R^2\frac{\partial T}{\partial t}$$

$$\boxed{\dot{q}A - Ph(T-T_\infty) - kA\frac{\partial^2 T}{\partial x^2} = \rho C_p A\frac{\partial T}{\partial t}}, \quad \boxed{A = \pi R^2 \quad P = 2\pi R}$$

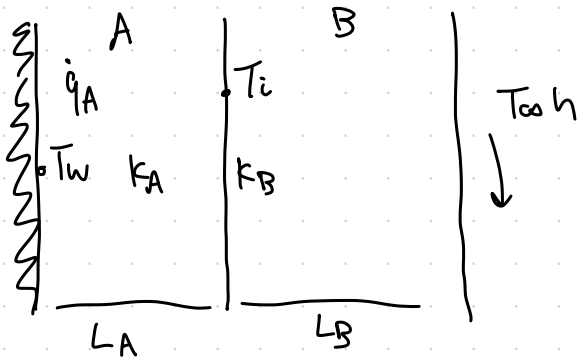
c) $T(x=0) = T_1$ $\quad -k\frac{dT}{dx}\Big|_{x=L} = h(T(L)-T_\infty)$

$$\boxed{\begin{aligned} &\text{IC: } T(x,0) = T_i \\ &\text{BC: } T(0,t) = T_1 \quad -k\frac{dT}{dx}\Big|_{x=L} = h(T(L,t)-T_\infty) \end{aligned}}$$

Given: $L_A = 30cm$ $\dot{q}_A$ k_A $L_B = 10cm$ $k_B = 30 \frac{W}{mK}$ $\dot{q}_B = 0$ $T_\infty = 20°C$ $h = 500 \frac{W}{m^2K}$
$T_w = 115°C$ $T_i = 100°C$

Find a) T_s b) $\dot{q}$ c) k_A d) Plot

Schematic:



Assume: 1D, left A insulated, no contact resistance

Analysis:

a) $q''_{cond} = q''_{conv}$ $\quad k_B \frac{T_i - T_s}{L_B} = h(T_s - T_\infty)$

$30 \frac{W}{mK}\left(\frac{100C - T_s}{.1m}\right) = 500 \frac{W}{m^2K}(T_s - 20C)$ $\boxed{T_s = 50°C}$

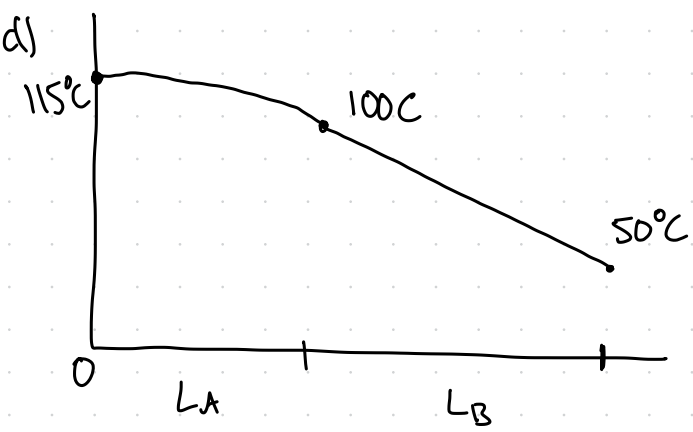
b) $q'' = \dot{q} L_A$ $\quad q'' = k_A\left(\frac{T_w - T_i}{L_A}\right) = h(T_s - T_\infty) = 15000 \frac{W}{m^2}$

$\dot{q} = 15 \frac{kW}{m^2} / .3m = \boxed{50 \frac{kW}{m^3}}$

c) $T(x) = T_w - \frac{\dot{q}x^2}{2k_A}$ $\quad T(L_A) = 100°C$

$T_i = T_w - \frac{\dot{q}L_A^2}{2k_A}$ $\quad \boxed{k_A = 150 \frac{W}{mK}}$

d)



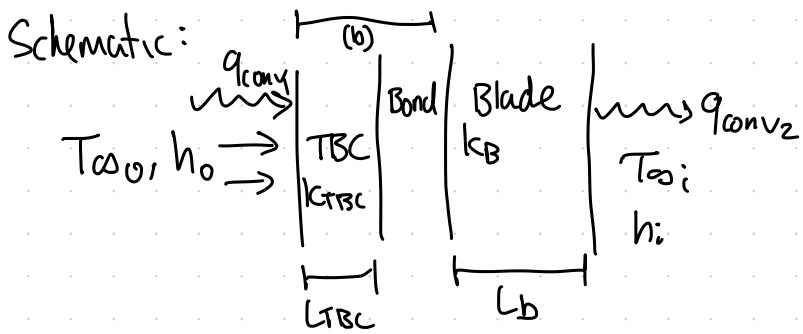
3) Given $k_b = 25 \frac{W}{mK}$ $k_{TBC} = 1 \frac{W}{mK}$ $R'' = 10^{-4} \frac{m^2K}{W}$ $T_{\infty,o} = 1616K$ $T_{\infty i} = 400K$

$h_o = 1000 \frac{W}{m^2K}$ $h_{air} = 500 \frac{W}{m^2K}$ $L_b = 5mm$ $T_{max} = 1200K$

Find a) thermal resistance network T_s no TBC

b) With TBC $L_{TBC} = .5mm$ TR network $T_s < T_{max}$?

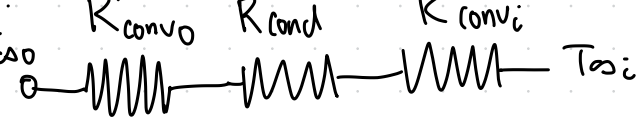
Schematic:



Assumptions: 1D No radiation, Steady State, uniform material

Analysis:

a)



$R''_{conv_o} = \frac{1}{h_o} = .001 \frac{m^2K}{W}$

$R''_{cond} = \frac{L_b}{k_b} = .0002 m^2 \frac{K}{W}$

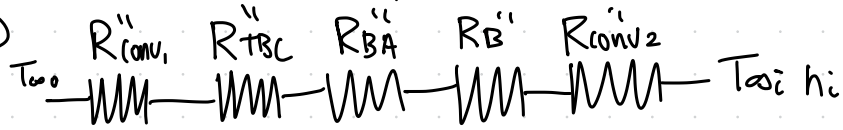
$R''_{tot} = R''_{conv_o} + R''_{cond} + R''_{conv_i} = .0032 \frac{m^2K}{W}$ $R''_{conv_i} = \frac{1}{h_i} = .002 m^2 \frac{K}{W}$

$$q'' = \frac{T_{\infty o} - T_{\infty i}}{R''_{tot}} = \frac{1616K - 400K}{.0032 \frac{Wm^2}{K}} = 380 \frac{kW}{m^2}$$

$$T_{so} = T_{\infty o} - (q'' R''_{conv_o}) = 1616K - 380 \frac{kW}{m^2}(.001) = \boxed{1236K}$$

Above the 1200K limit

b)



$R''_{conv_1} = .001 \frac{m^2K}{W}$ $R''_{TBC} = \frac{L_{TBC}}{k_{TBC}} = .0005 m^2 \frac{K}{W}$ $R''_{BA} = 10^{-4} \frac{m^2K}{W}$

$R''_B = .0002 m^2 \frac{K}{W}$ $R''_{conv_2} = .002 \frac{m^2K}{W}$

$$q'' = \frac{T_{\infty o} - T_{\infty i}}{R''_{tot}} = 320 \frac{kW}{m^2}$$

$$T_s = 1616K - \left(320 \frac{kW}{m^2}(.001 + .0005 + .0001)\right) = \boxed{1104K}$$

$1104 < 1200K$, effective solution